

# Joint Distributions

## Definition (Joint PMF)

The joint PMF of discrete r.v.s  $X$  and  $Y$  is the function  $p_{X,Y}$  given by

$$p_{X,Y}(x,y) = P(X = x, Y = y)$$

The joint pmf of  $X$  and  $Y$  should have the following properties.

(1)  $p_{X,Y}(x,y) \geq 0$  , (2)  $\sum_{x,y} p_{X,Y}(x,y) = 1$

**Eg 1.** A biased coin with  $P(H) = \frac{1}{3}$ , is tossed three times. Let

$X$  = number of head on the first toss

$Y$  = total number of heads

Find  $p_{X,Y}(x,y)$ .

$$P(X=0, Y=0) = \left(\frac{2}{3}\right)^3$$

$$P(X=1, Y=0) = 0$$

$$\begin{aligned} P(X=0, Y=1) &= \binom{2}{1} \left(\frac{1}{3}\right) \left(\frac{2}{3}\right)^2 \\ &= 2 \cdot \frac{1}{3} \cdot \frac{4}{9} = \frac{8}{27} \end{aligned}$$

|       | $Y=0$          | $Y=1$          | $Y=2$          | $Y=3$          |
|-------|----------------|----------------|----------------|----------------|
| $X=0$ | $\frac{8}{27}$ | $\frac{8}{27}$ | $\frac{2}{27}$ | 0              |
| $X=1$ | 0              | $\frac{4}{27}$ | $\frac{4}{27}$ | $\frac{1}{27}$ |

$$P(X=0, Y=2) = \binom{2}{2} \left(\frac{1}{3}\right)^2 \cdot \left(\frac{2}{3}\right) = \frac{2}{27}$$

$$P(X=0, Y=3) = 0$$

$$P(X=1, Y=1) = \binom{2}{0} \frac{1}{3} \left(\frac{2}{3}\right)^2 = \frac{4}{27}$$

$$P(X=1, Y=2) = \binom{2}{1} \left(\frac{1}{3}\right)^2 \frac{2}{3} = \frac{4}{27}$$

$$P(X=1, Y=3) = \binom{2}{2} \left(\frac{1}{3}\right)^3 = \frac{1}{27}$$

From the joint distribution of  $X$  and  $Y$ , we can get the distribution of  $X$  alone by summing over the possible values of  $Y$ . This gives us the familiar PMF of  $X$  that we have seen in previous chapters. In the context of joint distributions, we will call it the marginal distribution of  $X$ .

### Definition (Marginal PMF of $X$ )

For discrete r.v.s  $X$  and  $Y$ , the marginal PMF of  $X$  is



$$P(X = x) = \sum_y P(X = x, Y = y) \quad (\text{column sum})$$

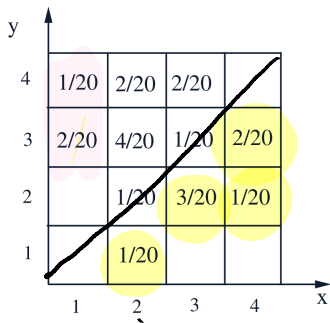
The marginal PMF of  $X$  is the PMF of  $X$ , viewing individually rather than jointly with  $Y$ .

**Note:** The marginal distribution of  $Y$  is defined analogously. i.e.,



$$P(Y = y) = \sum_x P(X = x, Y = y) \quad (\text{row sum})$$

Eg 2. (a)  $P(X=1)$  (b)  $P(X>Y)$



$$\begin{aligned}P(X=1) &= P(X=1, Y=3) \\ &\quad + P(X=1, Y=4) \\ &= \frac{2}{20} + \frac{1}{20} = \frac{3}{20}\end{aligned}$$

$$\begin{aligned}P(X>Y) &= P(X=2, Y=1) + P(X=3, Y=2) + P(X=4, Y=3) \\ &\quad + P(X=4, Y=2) \\ &= \frac{2}{20} + \frac{1}{20} + \frac{3}{20} + \frac{1}{20} = \frac{7}{20}\end{aligned}$$

## Definition (Joint Pdf)

The joint Pdf of two continuous r.v.s  $X$  and  $Y$  is the function  $f(x, y)$  where for any two dimensional set  $A$

$$P((x, y) \in A) = \int \int_A f(x, y) dx dy$$

The joint pdf of  $X$  and  $Y$  should have the following properties.

(1)  $f_{X,Y}(x, y) \geq 0$  , (2)  $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$

**Note:** For two random variables  $X$  and  $Y$ , we can state that “volume under the surface  $f(x, y)$  = probability for an  $xy$  region with area  $A$ ”

Eg 3-a: Two continuous random variables  $X$  and  $Y$  have joint probability density function

$$f(x, y) = \begin{cases} \frac{1}{2xy} & 1 \leq x \leq e, 1 \leq y \leq e^2 \\ 0 & \text{otherwise} \end{cases}$$

What is the probability that neither  $X$  nor  $Y$  is more than 2?

Recall neither  $A$  nor  $B$  means  $A^c \cap B^c$

$$A = X > 2$$

$$B = Y > 2$$

$$A^c \cap B^c = X \leq 2 \text{ and } Y \leq 2$$

$$P(A^c \cap B^c) = P(X \leq 2 \text{ and } Y \leq 2)$$

$$= \int_1^2 \int_1^2 \frac{1}{2xy} dx dy = \frac{1}{2} \left[ \int_1^2 \frac{dx}{x} \right] \left[ \int_1^2 \frac{dy}{y} \right]$$

$$= \frac{1}{2} (\ln|x|)_1^2 (\ln|y|)_1^2 = \frac{1}{2} (\ln 2)(\ln 2) = \frac{(\ln 2)^2}{2}$$

## Definition (Marginal PDF)

The marginal probability density functions of  $X$  and of  $Y$  are given by

$$f(x) = \int_{-\infty}^{\infty} f(x, y) dy, \quad f(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

**Eg 3-a:** Two continuous random variables  $X$  and  $Y$  have joint probability density function

$$f(x, y) = \begin{cases} \frac{1}{2xy} & 1 \leq x \leq e, 1 \leq y \leq e^2 \\ 0 & \text{otherwise} \end{cases}$$

Find the two marginal probability density functions.

when  $x < 1$  or  $x > e$

$$f(x) = \int_{-\infty}^{\infty} f(x, y) dy = 0$$



when  $1 < x < e$

$$\begin{aligned} f(x) &= \int_{-\infty}^{\infty} f(x,y) dy = \int_1^{e^2} \frac{dy}{2xy} = \frac{1}{2x} \int_1^{e^2} \frac{dy}{y} \\ &= \frac{1}{2x} \ln|y| \Big|_1^{e^2} = \frac{1}{2x} (\ln e^2 - \cancel{\ln 1}) \end{aligned}$$

$$f(x) = \frac{1}{2x} \cdot 2 = \frac{1}{x}$$

Hence

$$f(x) = \begin{cases} \frac{1}{x} & 1 \leq x \leq e \\ 0 & \text{otherwise} \end{cases}$$

when  $y < 1$  or  $y > e^2$

$$f(y) = \int_{-\infty}^{\infty} f(x, y) dx = 0$$

when  $1 < y < e^2$

$$f(y) = \int_{-\infty}^{\infty} f(x, y) dx = \int_1^e \frac{1}{2xy} dx$$

$$f(y) = \frac{1}{2y} \int_1^e \frac{dx}{x} = \frac{1}{2y} \ln|x| \Big|_1^e$$

$$f(y) = \frac{1}{2y} (\ln e - \cancel{\ln 1}) = \frac{1}{2y}$$

$$f(y) = \begin{cases} \frac{1}{2y} & 1 < y < e^2 \\ 0 & \text{otherwise} \end{cases}$$

discrete case  $E[g(X, Y)] = \sum_x \sum_y g(x, y) p_{X,Y}(x, y)$

continuous case  $E[g(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f(x, y) dx dy$

Eg 4.  $E[X+Y]$

|         |         |         |         |
|---------|---------|---------|---------|
| $x = 1$ | 0.15    | 0.1     | 0       |
| $x = 0$ | 0.25    | 0.3     | 0.2     |
|         | $y = 0$ | $y = 1$ | $y = 2$ |

Method 1:

$$E[X] = \sum_x x P_X(x)$$

$$\begin{aligned} &= 0 \cdot P_X(0) + 1 \cdot P_X(1) \\ &= 0 \cdot (0.75) + 1 \cdot (0.25) \\ &= 0.25 \end{aligned}$$

|       |       |       |       |      |
|-------|-------|-------|-------|------|
| ↑ sum | 0.4   | 0.4   | 0.2   | sum→ |
| x = 1 | 0.15  | 0.1   | 0     | 0.25 |
| x = 0 | 0.25  | 0.3   | 0.2   | 0.75 |
|       | y = 0 | y = 1 | y = 2 |      |

$$E[Y] = \sum_y y P_Y(y) = 0 \cdot P_Y(0) + 1 \cdot P_Y(1) + 2 \cdot P_Y(2)$$

$$= 0 \cdot (0.4) + 1 \cdot (0.4) + 2 \cdot (0.2) = 0.4 + 0.4 = 0.8$$

$$E[X + Y] = E[X] + E[Y] = 0.8 + 0.25 = 1.05$$

Method 2:

$$E[X+Y] = \sum_{(x,y)} (x+y) P(X=x, Y=y)$$

$$E[X+Y] = (0+0) P(X=0, Y=0) + (0+1) P(X=0, Y=1)$$

$$+ (0+2) P(X=0, Y=2) + (1+0) P(X=1, Y=0)$$

$$+ (1+1) P(X=1, Y=1) + (1+2) P(X=1, Y=2)$$

$$= P(X=0, Y=1) + 2 P(X=0, Y=2) + P(X=1, Y=0)$$

$$+ 2 P(X=1, Y=1) + 3 P(X=1, Y=2)$$

$$= 0.3 + 2(0.2) + 0.15 + 0.2 + 3(0) =$$

$$= 0.3 + 0.4 + 0.15 + 0.2 = 0.9 + 0.15 = 1.05$$